

Project Title and Faculty Mentor

The Voice of Code: Optimizing Accessibility in Web-Based Educational Tools through Real-Time Speech-to-Text Integration.

Faculty Mentor: Dr. Paul Gazzillo, UCF Department of Computer Science.

Project Objective or Aim

The objective of this research is to investigate the effectiveness of integrating the Web Speech API into web-based educational environments to support students with motor impairments. Specifically, this study aims to measure how voice-activated command recognition influences the speed and accuracy of web application debugging. By developing a specialized interface, the research will answer: To what extent does a customized technical voice-lexicon reduce the "barrier to entry" for students who cannot utilize traditional keyboard and mouse inputs? The project seeks to establish a framework for hands-free navigation in complex technical interfaces, ensuring that coding remains an accessible skill regardless of physical ability.

Project Background and Significance

Digital accessibility is a keystone of inclusive education, yet specialized fields like computer science often lack the necessary tools to support students with motor disabilities. Standard assistive technologies often fail when encountering technical syntax; for example, a generic speech-to-text tool may not distinguish between the word "bracket" and the character "{", leading to high error rates in debugging tasks. This syntax gap effectively bars many capable students from pursuing technical degrees.

This project is grounded in the Universal Design for Learning (UDL) framework, which argues that educational materials should provide multiple means of engagement and expression. By leveraging the Web Speech API, we can create a solution that works directly in the browser, making accessibility tools more portable and less stigmatizing for students. Furthermore, this research builds on the Human-Computer Interaction (HCI) theory of "Invisible Technology," where the interface becomes a seamless extension of the user's intent.

The significance of this study extends to the UCF community and the broader STEM field. As UCF continues to expand its online and hybrid engineering programs, providing validated, accessible alternatives for lab work is essential for compliance and equity. This research will provide the empirical data needed to justify the integration of voice-command structures in future web-based IDEs and educational platforms, ultimately fostering a more diverse and inclusive technical workforce.

Research Methods

This project will follow a three-phase development and testing cycle during the Fall 2026 semester:

- Phase 1: Development (Weeks 1-5): I will design a web-based interface using JavaScript and React that utilizes the Web Speech API. The focus will be on creating a custom library of voice triggers specifically for debugging actions (e.g., "step over," "inspect variable," "console log").
- Phase 2: User Testing (Weeks 6-10): I will recruit 10-15 student volunteers to perform a series of standardized debugging tasks. Half the group will use traditional inputs, while the other half uses the voice-integrated interface. I will collect data on "Time to Completion" and "Error Frequency."
- Phase 3: Analysis and Reporting (Weeks 11-15): Data will be analyzed using comparative statistics to determine the efficiency gain of the voice interface.

Each step is necessary to ensure the tool is not just functional, but demonstrably better for the target demographic than current solutions.

Expected Outcome

The primary deliverable for this project will be a White Paper detailing the interface architecture and the results of the user testing phase. This document will be shared with the UCF Student Accessibility Services (SAS) to provide a roadmap for implementing similar technologies in university labs. Additionally, the project will culminate in a poster presentation for the UCF Showcase of Undergraduate Research Excellence (SURE).

The new insight gained from this project will be a clearer understanding of "Voice-Syntax Mapping", specifically, how developers can map natural language to complex code structures without increasing cognitive load. For the UCF community, this project will serve as a proof-of-concept for low-cost, browser-based accessibility tools that can be implemented across various departments without requiring expensive hardware upgrades.

Literature Review

1. Bigham, J. P., et al. (2010). "WebAnywhere: A Non-Visual Interface to the Graphical Web." Proceedings of the ASSETS Conference.
2. Shneiderman, B., & Plaisant, C. (2010). Designing the User Interface: Strategies for Effective Human-Computer Interaction. Pearson.
3. W3C (2021). "Web Speech API Specification." World Wide Web Consortium.
4. Stefik, A., & Siebert, S. (2013). "An Empirical Investigation into Programming Language Syntax." ACM Transactions on Computing Education.
5. Burgstahler, S. (2015). Universal Design in Higher Education: From Principles to Practice. Harvard Education Press.

Preliminary Work and Experience

I have prior experience as a web application debugger, which has given me a deep understanding of the common bottlenecks in software troubleshooting. My academic background in computer science has trained me with a strong foundation in Data Structures and Algorithms (DSA) and Object-Oriented Programming (OOP), both of which are essential for developing the logic behind the proposed interface. Furthermore, I have already conducted successful preliminary tests using the Web Speech API to detect specific triggers in a browser environment. This demonstrates my technical capacity to implement the complex command-and-control structures required for this research.

IRB/IACUC Statement

This proposal requires IRB approval. Because the research involves testing the interface with human subjects to measure navigation speed and error rates, approval from the Institutional Review Board is necessary to ensure the ethical treatment and privacy of all participants.

Budget

Item	Description	Cost
Hosting & Domain	1 year of AWS/Vercel hosting for the testing interface	\$150.00
Testing Hardware	3 High-fidelity noise-canceling microphones for subjects	\$450.00
Participant Incentives	\$20 Gift cards for 15 student testers	\$300.00
Conference Fees	Printing costs for SURE poster and materials	\$100.00
Total		\$1,000.00